Variant Detection in Brassica Napus

University of York
MRes Project
Katarina Gmeiner

Presentation Outline

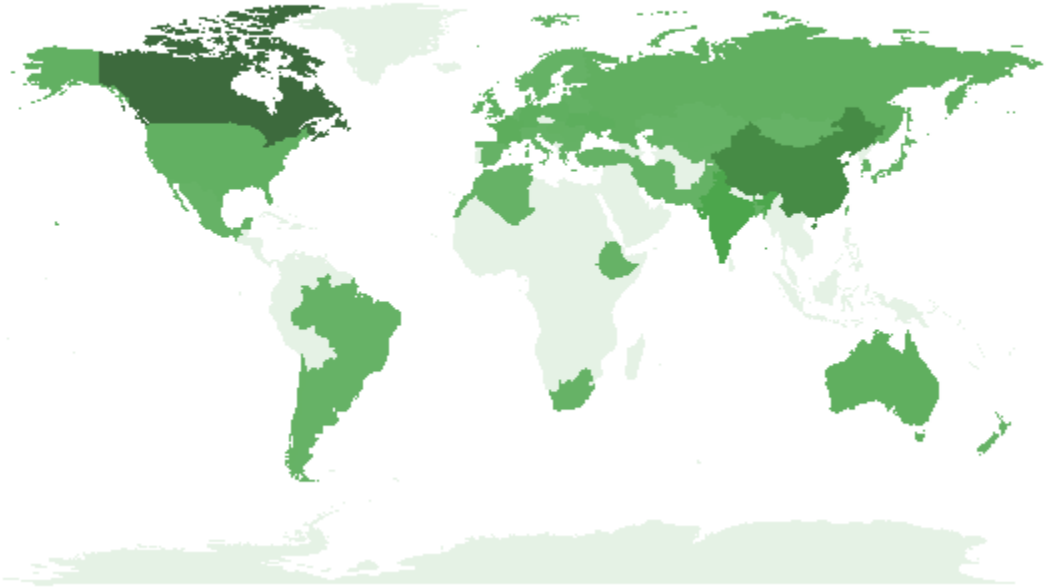
1. Introduction
2. Computational Approach
3. Results

Introduction

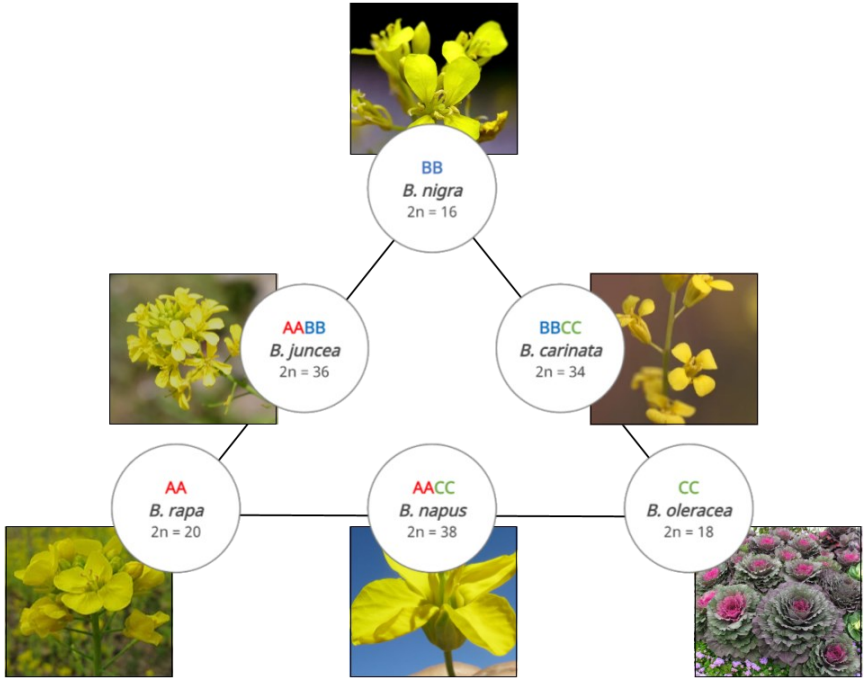
Introduction

Rapeseed Production By Country (2019)

Source: FAOSTAT, http://www.fao.org/faostat/en/#data/QC/



Triangle of U



Introduction

- Advancements in NGS technology lead to improvements in DNA Sequencing
- Molecular Markers became more easily available
- Single Nucleotide Polymorphisms (SNPs) most common marker
- Gamma Radiation Panels used to induce mutations

Genes of Interest

FAD2

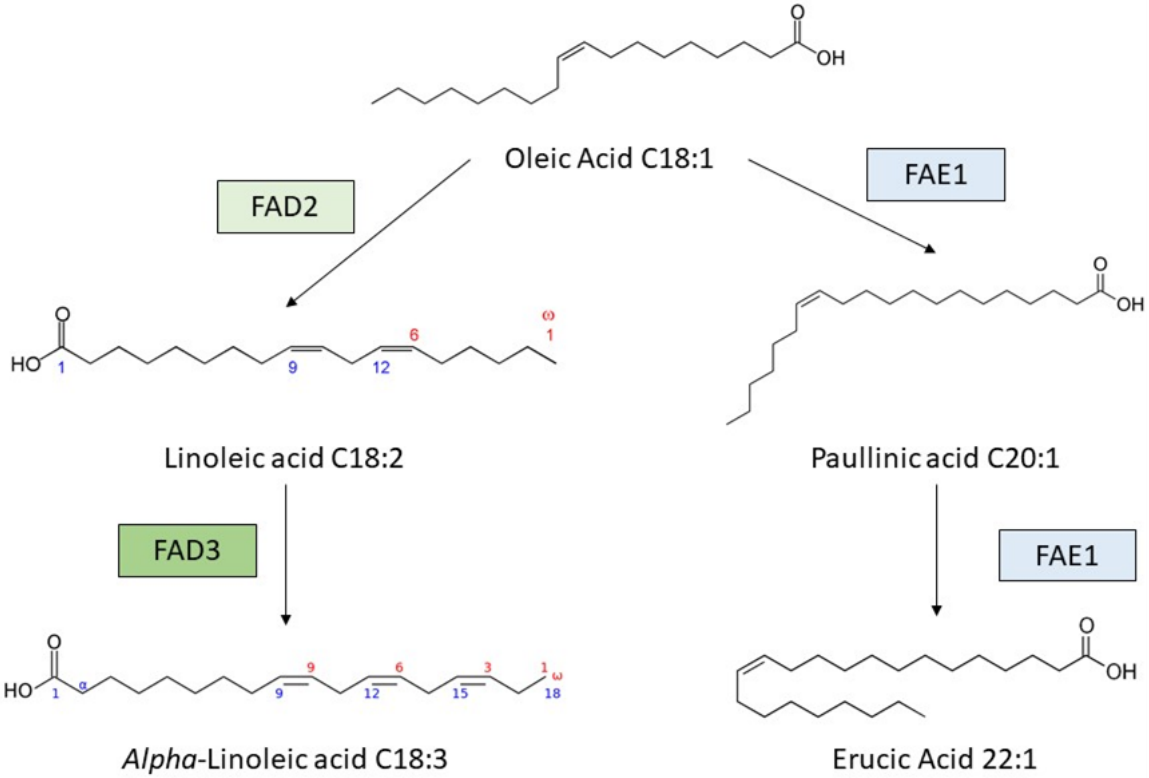
- Oleic Acid → Linoleic Acid
- 4 orthologues in *B. napus*

FAE1

- Key gene for erucic acid biosynthesis
- 2 genes in *B. napus*

Genetic variants from both genes known to affect FA content

FAE1 - FAD2 Pathway



Challenges

- Variant detection in polyploid organisms more complicated
- Many programmes are designed for diploid organisms (humans)
- Some variants are harder to detect
- INDEL detection

Computational Analysis

FASTA and FASTQ

/mnt/lustre/groups/biol-pgen-2019/kg913/Brassica/results/Bo3g168810.1/f...

Zeichenkodierung Farbe

```
>C03:64359517-64361037
TTAGGACCGACCGTTTTGGACACGAGTCTCTGACTTACCTGAATCAGAATCAATTTTAAC
CGGRTATCTGTCGATGCAGTGTTCCCAAGGACTATTTGTYGAAGCTTTGACATTGTTTAG
AGCCACCCAAACTGCACTGTTACACTTAAAGCCTGACCCTAAAGCAATCTGCCAAACTTT
ATTACCTTTCTTCATCCTTCCTTTTGCTTCTATGTATGCCAACTCATACCATATTGAGCT
AGATGAAGTGTTTCCAAATCTATGTAACGTTGATCTTGATGCCTCTACATCGATCGGTGC
TAGGCCTAGGTTCTTCTCTAGCACATCAATCACGGCTTTGCCTCCGGCATGTATACAAAA
ATGGTCGATAGCAAGCTTGAAGTCCGGGACGTAATAATGTTTGATTTTGTCTTTGAAAAG
TTTCTTGCCCATGAAGGTAACGGAAAAAAGAAGTTTCTCGCTTAACGGAAGAATCAACGG
ACCCAGCGTTGCTATGTTTTTCTTAACCGTTCGACCAGCAACATCGGTTATGTCCTTGGA
CAAACTCACTCCGGTTTTGCCGTTCTCATCGTCTCCTTGTTGCACGCAACGAAAAGACTT
GTCGTCAGCTCCGGTATGCGTTCGAACCGTGTGAACTAGCTCGTACTTGGACCGTCTACG
ATCTCTAGGCTTGTTGGAGAGCAAAATAGCGGCCCCACCAACACGGAACAAGCAATTTGA
AACCATCATGGACCTATTATCACCAGCGTAAATGTTATAAGTGATGTTCTCTGTGCTCAC
CACAAGAGCATACGTATTTTTATGGACATGCAACAAGTCCTTTGCTAGATCAATGGCTAT
AACGCCGGCACTACAACCCATGCCACCAAGGTTAAAGCTTCTTACGTTGCTTCGGAGCTT
GAAAGTGTTAACGACCATCGCGGAGAGCGAAGGAGTTGGATTAAACATGCTTGAGTTCAC
CACAAGTATACCTATATCTGTAGGGTTAACATTGGTGTTCTTGAATAGATTTTCTAGCGC
ACCAATGATAACCTGCTCCGTCTCTTCACGCGCCGCCGCAAAAGTCTTCCGGGGAGGGAC
CTGAAGCAGCCCCTCGGGCCCGTGGGTTTCATCGCCTAGACCTGAACGTTCTTGAATCTT
CCTCAAGAAGTCAAGCCAGGACGAGTCATCGCACGTGCCGTTCCGAGAAGGATCAGCTTT
TCTTACTTGGTAAAAGATATCCATGACCTTGGAGATACTTGATCTACAATGCGTTGGTGG
AAGGTAGCATGAGTACTCAACGAGGTAAACCGGTTTGGGCCGGGTTGCGATGTAGAGAAC
CGAACCGAAAACGGTGAAGGCAAAGAGTGGAGCAATGGTTATGAGGTTGTGTTGGAGATA
GGAATAGTATAAGTGGTGAAGATCGTCTATGGTAAGCCGATAGGCTTTTCCGGCGACGAT
CGCCGTTAATGGAAAGAAACAAAGGTTGAAAAGGTTGGTTATGACGTAATGGTAAAGGAG
CTTTACGTTAATGGACGTCAT
```

Zeile: 1/27 Zeichen: 62 (0x3E) Kodierung: 1252 (ANSI - L

/mnt/lustre/groups/biol-pgen-2019/kg913/Brassica/results/Bo3g168810.1/fa...

Zeichenkodierung Farbe

```
@E00499:80:HCNGTALXX:5:2104:12246:42815/1
AAGAGATACATACATGTAAAACAATCAAATACTAGAACACCGTTGCATTCTTTATTCATAACAAACAAGAC
+
AAFFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ
@E00499:80:HCNGTALXX:5:1222:23328:59587/2
AAAACAATCAAATACTAGAACACCGTTGCATTCTTTATTCATAACAAACAAGAGAAACATCGTAGCCATCA
+
AAFFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ
@E00499:80:HCNGTALXX:5:2221:23449:50480/2
CAAATACTAGAACACCGTTGCATTCTTTATTCATAACAAACAAGAGAAACATCGTAGCCATCAAATTATTA
+
AAAFFJJJFFJJJJJJJJJJJJJJJFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJJJJJJJJJJJ
@E00517:26:HCY35ALXX:6:2123:3813:4684/1
GTTAAAATTGATTCTGATTCAGGTAAGTCAGAGACTCGTGTCCAAAACGGTCGGTCCTAATAAATGATGTT
+
AAFFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJFJJJJJAJJJJJJJJJJJJJJJJJJJFJJJJJFJJFJ
@E00517:72:HCNNYALXX:1:1122:29386:47580/1
TATTCATAACAAACAAGAGAAACATCGTAGCCATCAAATTATTATAACCAATAAAAAGAAACGAAAGAGAC
+
<-A-FFJJJJJJJJJFFF7JJJJFJFAA-AF7FAJJFJJFFAFF7JJJJ-FJAJJF-JJFFFJJJFJAJ<
@E00516:39:HC5LVALXX:5:1123:25286:45013/1
AAAGAAACGAAAGAGAGCAAACATCATTTATTAGGACCGACCGTTTTGGACACGAGTCTCTGACTTACCTC
+
AAFFFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ
@E00499:80:HCNGTALXX:5:1205:30025:59341/2
TGGGTGGCTCTAAACAATGTCAAAGCTTCAACAAATAGTCCTTGGGAACACTGCATCGACAGATACCCGGT
+
<AA-AFJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJJ
@E00499:80:HCNGTALXX:5:1205:29985:59552/2
TGGGTGGCTCTAAACAATGTCAAAGCTTCAACAAATAGTCCTTGGGAACACTGCATCGACAGATACCCGGT
+
-A--<-AF-AFJFJF<AJAA<FFFJFJ7FJJFJJJJFJFA<7FJ<A<AFFJJJJJF<AF-<7FFJ<AJA7
```

Zeile: 1/316 Zeichen: 64 (0x40) Kodierung: 1252 (ANSI - L

Binary Alignment/Map (BAM)

kg913@login2:~/scratch/bio/Gamma/bam/Radiation

```
         25129381  25129391  25129401  25129411  25129421  25129431  2512
GCTGTCCAAGGAGTTGCCTCGATGGTCTGCTTCTACGGAGTCCCGCTTCTGATAGTCAACGGGTTCTTAGTT
R.......................................T..Y........T....................
............                   ,,,,,,,,,,t,,t,,,,,,,,t,,,,,,,,,,,,,,,,,,,
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A.......................................T..T........T....................
A.......................................T..T........T....................
A.......................................T..C........T....................
         ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,t,,t,,,,,,,,t,,,,,,,,,,,,,,,,,,,
         ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,t,,t,,,,,,,,t,,,,,,,,,,,,,,,,,,,
         ,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,,t,,c,,,,,,,,t,,,,,,,,,,,,,,,,,,,
                                  .....T..T........T....................
```

samtools tview FNT80-6.sorted.bam ../../reference/ref.fa -p A05:25129372

/mnt/lustre/groups/biol-pgen-2019/kg913/Brassica/results/Bo3g168810.1/vcf/Radiation-SNP.vcf - Viking - Textbearbeitungsprogamm - WinSCP

Zeichenkodierung Farbe

```
##contig=<ID=BnaC03g57890.1T,length=681>
##contig=<ID=BnaC03g57900.1T,length=399>
##contig=<ID=BnaC03g57910.1T,length=702>
##contig=<ID=BnaC03g49430.1T,length=1071>
##ALT=<ID=*,Description="Represents allele(s) other than observed.">
##INFO=<ID=INDEL,Number=0,Type=Flag,Description="Indicates that the variant is an INDEL.">
##INFO=<ID=IDV,Number=1,Type=Integer,Description="Maximum number of raw reads supporting an indel">
##INFO=<ID=IMF,Number=1,Type=Float,Description="Maximum fraction of raw reads supporting an indel">
##INFO=<ID=DP,Number=1,Type=Integer,Description="Raw read depth">
##INFO=<ID=VDB,Number=1,Type=Float,Description="Variant Distance Bias for filtering splice-site artefacts in RNA-seq data (bigger is better)",Version="3">
##INFO=<ID=RPB,Number=1,Type=Float,Description="Mann-Whitney U test of Read Position Bias (bigger is better)">
##INFO=<ID=MQB,Number=1,Type=Float,Description="Mann-Whitney U test of Mapping Quality Bias (bigger is better)">
##INFO=<ID=BQB,Number=1,Type=Float,Description="Mann-Whitney U test of Base Quality Bias (bigger is better)">
##INFO=<ID=MQSB,Number=1,Type=Float,Description="Mann-Whitney U test of Mapping Quality vs Strand Bias (bigger is better)">
##INFO=<ID=SGB,Number=1,Type=Float,Description="Segregation based metric.">
##INFO=<ID=MQ0F,Number=1,Type=Float,Description="Fraction of MQ0 reads (smaller is better)">
##FORMAT=<ID=PL,Number=G,Type=Integer,Description="List of Phred-scaled genotype likelihoods">
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##INFO=<ID=ICB,Number=1,Type=Float,Description="Inbreeding Coefficient Binomial test (bigger is better)">
##INFO=<ID=HOB,Number=1,Type=Float,Description="Bias in the number of HOMs number (smaller is better)">
##INFO=<ID=AC,Number=A,Type=Integer,Description="Allele count in genotypes for each ALT allele, in the same order as listed">
##INFO=<ID=AN,Number=1,Type=Integer,Description="Total number of alleles in called genotypes">
##INFO=<ID=DP4,Number=4,Type=Integer,Description="Number of high-quality ref-forward , ref-reverse, alt-forward and alt-reverse bases">
##INFO=<ID=MQ,Number=1,Type=Integer,Description="Average mapping quality">
##bcftools_callVersion=1.10.2+htslib-1.10.2
##bcftools_callCommand=call -mv; Date=Sun Jul 25 13:03:16 2021
#CHROM  POS     ID      REF     ALT     QUAL    FILTER  INFO    FORMAT  ../../../../results/Bo3g168810.1/Radiation/bam/FNT100_1a.sub.sort.bam   ../../../../results/Bo3g168810.1/Radiation/bar
        ../../../../results/Bo3g168810.1/Radiation/bam/G2000-14a.sub.sort.bam   ../../../../results/Bo3g168810.1/Radiation/bam/G2000-150a.sub.sort.bam  ../../../../results/Bo3g168810.1/Radia
        ../../../../results/Bo3g168810.1/Radiation/bam/g2000-347-2.sub.sort.bam ../../../../results/Bo3g168810.1/Radiation/bam/g2000-348-2.sub.sort.bam ../../../../results/Bo3g168810.1/Radia
C03     64359481        .       A       C       999     .       DP=3907;VDB=9.07588e-11;SGB=-148.582;RPB=0.274492;MQB=5.59632e-11;MQSB=5.64983e-28;BQB=0.943651;MQ0F=0.203481;AC=1146;AN=1146
C03     64359580        .       A       G       999     .       DP=3623;VDB=1.74566e-05;SGB=-124.755;RPB=1;MQB=1;MQSB=0;BQB=1;MQ0F=0.303616;AC=1134;AN=1134;DP4=0,1,1307,1702;MQ=17     GT:PL
C03     64359616        .       C       T       999     .       DP=3505;VDB=0.998506;SGB=-101.491;RPB=4.61128e-23;MQB=1.80241e-16;MQSB=0;BQB=0.184328;MQ0F=0.376605;AC=1122;AN=1122;DP4=61,15
C03     64359959        .       G       A       572     .       DP=50;VDB=0.000320742;SGB=1.42575;RPB=1;MQB=1;MQSB=0.304348;BQB=1;MQ0F=0.04;ICB=0.0023215;HOB=0.00113379;AC=41;AN=42;DP4=1,0,
C03     64360549        .       C       T       435     .       DP=41;VDB=0.0590036;SGB=2.03991;RPB=1;MQB=1;MQSB=0.0117219;BQB=1;MQ0F=0.195122;AC=42;AN=42;DP4=0,1,20,5;MQ=18   GT:PL   ./.:0
C03     64360686        .       C       T       3.02059 .       DP=2754;VDB=0.013828;SGB=-1.07924;RPB=0.97189;MQB=0.334113;MQSB=0.0688483;BQB=0.662441;MQ0F=0.748003;ICB=1;HOB=1.58306e-06;AC=
C03     64360692        .       T       C       30.9256 .       DP=2900;VDB=0.188618;SGB=3.07387;RPB=0.580458;MQB=0.157493;MQSB=0.0673044;BQB=0.0157281;MQ0F=0.741379;ICB=1;HOB=1.59439e-06;A
C03     64360749        .       A       G       7.36144 .       DP=4331;VDB=0.931964;SGB=-9.62037;RPB=0.882315;MQB=0.844033;MQSB=2.06797e-24;BQB=0.279691;MQ0F=0.616486;ICB=1;HOB=1.48121e-06
C03     64360817        .       C       T       24.9394 .       DP=5609;VDB=0.753018;SGB=-16.2554;RPB=0.653912;MQB=0.992406;MQSB=0;BQB=0.999978;MQ0F=0.487431;ICB=1;HOB=1.47107e-06;AC=1;AN=1
C03     64361015        .       A       G       7.62399 .       DP=8598;VDB=0.0766355;SGB=-74.6228;RPB=0.583107;MQB=0.988271;MQSB=0.723804;BQB=0.17574;MQ0F=0.0405908;ICB=1;HOB=1.47107e-06;A
C03     64361048        .       G       T       9.81713 .       DP=8086;VDB=0.00108242;SGB=-13.8861;RPB=0.669634;MQB=0.875589;MQSB=0.842266;BQB=0.00260337;MQ0F=0.000618353;ICB=1;HOB=1.47107
C03     64361129        .       T       G       8.11037 .       DP=3139;VDB=0.0270423;SGB=-9.62037;RPB=0.769153;MQB=0.998612;MQSB=0.728996;BQB=0.0110188;MQ0F=0.000637146;ICB=1;HOB=1.54435e-
```

Zeile: 1/4469 Spalte: 1 Zeichen: 35 (0x23) Kodierung: 1252 (ANSI - L

kg913@login2:~

```
login as: kg913
kg913@viking.york.ac.uk's password:
ast login: Wed Aug 11 12:04:26 2021 from 172.18.68.140
 'o`
'ooo`                                   Welcome to viking
`oooo`
 `oooo`         'o`                     Flight Direct 2018.3
   `ooooo`  `ooooo`              Based on CentOS Linux 7 (Core)
      `oooo:oooo`
          `v  -[ alces flight ]-
ocumentation on using Viking can be found at:
ttps://wiki.york.ac.uk/x/e4gKDQ

his document
ues you are

o submit sof
aving with V

kg913@login2
```

RStudio

File Edit Code View Plots Session Build Debug Profile Tools Help

Go to file/function Addins

ring.R | tree.a8 | vcf-analysis.R | gatk-script.R | submit-script.R | bam-script.R | bam3-script.R | vcf-script.R | exon2-script.R | exon-script.R

Source on Save Run Source

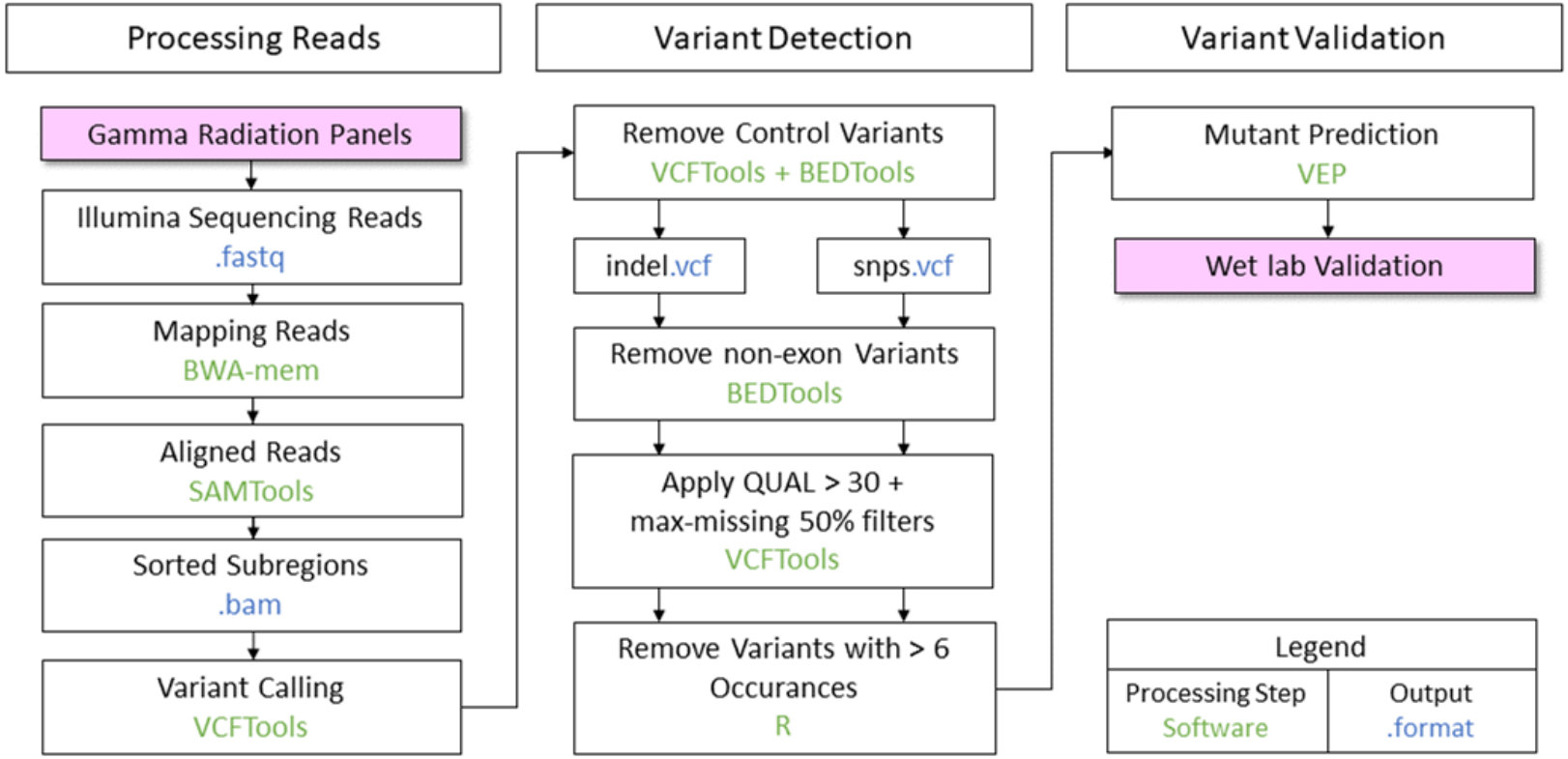
```
157
158  ## loop for bam files
159
160  for(i in 1:length(exp.condition)){
161    command1 <- paste(ID,"/",type, "/sh/", exp.condition[i], "-", ID, ".sh", sep="")
162    log.file <- paste("../log/", exp.condition[i], "-", ID, ".log", sep="")
```

/mnt/lustre/groups/biol-pgen-2019/kg913/Brassica/scripts/Bo5g134730.1/Radiation/sh/FNT40_1a-Bo5g134730.1.sh - Viking - Textbearbeitungsprogamm - WinSCP

Zeichenkodierung Farbe

```
#!/bin/bash
#SBATCH --job-name=Bo5g134730.1/Radiation/sh/FNT40_1a-Bo5g134730.1.sh
#SBATCH --mail-type=NONE
#SBATCH --mail-user=kg913@york.ac.uk
#SBATCH --ntasks=16
#SBATCH --mem=10gb
#SBATCH --time=11:59:00
#SBATCH --output=../log/FNT40_1a-Bo5g134730.1.log
#SBATCH --account=biol-pgen-2019
date
module load bio/SAMtools

  module load bio/SeqKit

  module load bio/BCFtools/1.10.2-GCC-9.3.0

  module load bio/VCFtools/0.1.15-foss-2018b-Perl-5.26.1

  module load lang/R
samtools view -h ../../../../../../Gamma/bam/Radiation/FNT40_1a.sorted.bam C05:44264985-44267481 > ../../../../results/Bo5g134730.1/Radiation/bam/FNT40_1a.su
samtools mpileup -uf ../../../../../../Gamma/reference/ref.fa ../../../../results/Bo5g134730.1/Radiation/bam/FNT40_1a.sub.sort.bam | bcftools call -c | vcfut
samtools bam2fq ../../../../results/Bo5g134730.1/Radiation/bam/FNT40_1a.sub.sort.bam > ../../../../results/Bo5g134730.1/Radiation/fastq/FNT40_1a-mapped.fastc
bcftools mpileup -f ../../../../../../Gamma/reference/ref.fa ../../../../results/Bo5g134730.1/Radiation/bam/*.bam| bcftools call -mv > ../../../../results/Bc
samtools faidx ../../../../../../Gamma/reference/ref.fa C05:44264985-44267481 > ../../../../results/Bo5g134730.1/Radiation/fasta/Bo5g134730.1-Radiation-refer
bcftools mpileup -f ../../../../../../Gamma/reference/ref.fa ../../../../results/Bo5g134730.1/Radiation/bam/FNT40_1a.sub.sort.bam| bcftools call -mv -O z > .
 tabix -p vcf ../../../../results/Bo5g134730.1/Radiation/fasta/FNT40_1a.vcf.gz
 samtools faidx ../../../../../../Gamma/reference/ref.fa C05:44264985-44267481 | bcftools consensus -I ../../../../results/Bo5g134730.1/Radiation/fasta/FNT46
cd ../../../../results/Bo5g134730.1/Radiation/vcf
vcftools --vcf Bo5g134730.1-Radiation.vcf --keep-only-indels --recode --recode-INFO-all --out Radiation-INDEL.vcf
vcftools --vcf Bo5g134730.1-Radiation.vcf --remove-indels --recode --recode-INFO-all --out Radiation-SNP.vcf
cp Radiation-INDEL.vcf.recode.vcf ../../vcf/Radiation-INDEL.vcf
cp Radiation-SNP.vcf.recode.vcf ../../vcf/Radiation-SNP.vcf
```

Methods


Processing Reads
Variant Detection
Variant Validation
Gamma Radiation Panels
Illumina Sequencing Reads
.fastq
Mapping Reads
BWA-mem
Aligned Reads
SAMTools
Sorted Subregions
.bam
Variant Calling
VCFTools
Remove Control Variants
VCFTools + BEDTools
indel.vcf
snps.vcf
Remove non-exon Variants
BEDTools
Apply QUAL > 30 +
max-missing 50% filters
VCFTools
Remove Variants with > 6
Occurances
R
Mutant Prediction
VEP
Wet lab Validation
Legend
Processing Step
Software
Output
.format

Overview Results

	Bo1g139590.1 FAD2.C1	Bo3g168810.1 FAE1.C3	Bo5g134730.1 FAD2.C5	Cab023705.1 FAD2.A5	Cab035983.1 FAE1.A8	Cab045628.2 FAD2.A1
Raw	44 SNPs 2 INDELs	14 SNPs 2 INDELs	104 SNPs 5 INDELs	62 SNPs 3 INDELs	40 SNPs 0 INDELs	40 SNPs 4 INDELs
Controls removed	42 SNPs 1 INDEL	8 SNPs 2 INDELs	96 SNPs 3 INDELs	47 SNPs 2 INDELs	35 SNPs	33 SNPs 2 INDELs
Exon only	42 SNPs 1 INDEL	6 SNPs 2 INDELs	36 SNPs 0 INDELs	9 SNPs 0 INDELs	13 SNPs	19 SNPs 1 INDEL
Filters	✖	1 SNP 2 INDELs	9 SNPs	12 SNPs	5 SNPs	6 SNPs 0 INDELs
<6 only	✖	1 SNP 0 INDELs	7 SNPs	11 SNPs	3 SNPs	✖

22 SNPs remaining for further analysis

Variation in FAD2

Variation in FAE1

Thank you for your attention! :)